Bhaskaracharya College of Applied Science

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Fundamental of Genetics report file

Topic: Eugenics

Can selective reproduction ever be scientifically or ethically justified?

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a. Introduction

Concept, scientific background, and relevance

What is eugenics?

An attempt to change the inherited features of a population by influencing who reproduces.

It has often involved ideas about “desirable” and “undesirable” people.

Scientific background

Traits can be inherited through genes. Francis Galton introduced the term eugenics.

But many traits also depend on environment and cannot be predicted by one gene.

Why is it relevant?

Genetic testing can help families understand inherited disease risks.

It also raises questions about choice, disability, fairness, and pressure on parents.

Key question: When does reducing disease risk become control over reproduction?

a. A short history

Why the past matters

1. **Galton’s ideas**

   He proposed encouraging the reproduction of people considered to have desirable inherited traits.

2. **Forced sterilization**

   Laws in the United States and other countries stopped some people from having children without free consent.

3. **Nazi Germany**

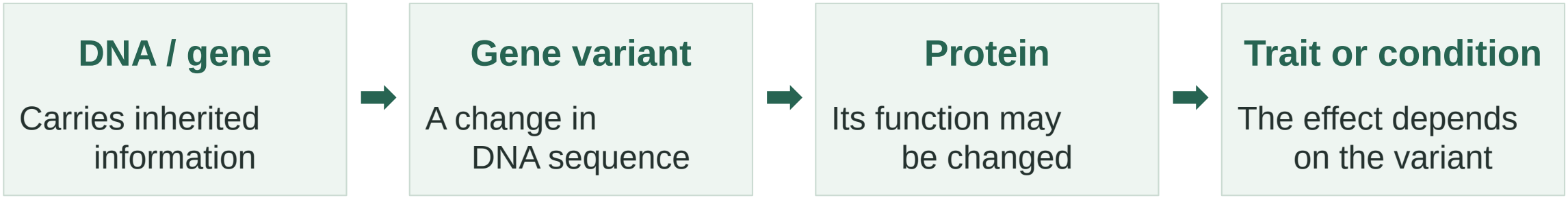
   Racial ideas and claims about inherited “defects” were used to support compulsory sterilization and persecution.

Lesson: Scientific language cannot justify discrimination or forced reproduction decisions.

b. Scientific basis: genes and molecules

How inherited differences can affect the body

DNA contains genes. Gene variants can change how proteins work and influence a trait or disease.



Example: Tay–Sachs disease

Some harmful variants in HEXA reduce the activity of the enzyme beta-hexosaminidase A. GM2 ganglioside then builds up in nerve cells and causes damage.

Many common conditions involve many genes plus environment, not one simple pathway.

b. Mechanism: a simple inheritance example

Two carriers of the same autosomal-recessive condition

Parent 1: A / a \ Parent 2: A / a	A	a
A	**AA** Without this variant	**Aa** Carrier
a	**Aa** Carrier	**aa** Affected genotype

For each conception

25%: affected genotype

50%: carrier

25%: without this variant

A = reference allele
a = pathogenic allele in this example

These are probabilities, not a fixed result for every four children.

b. Modern genetic technologies

Testing, selection, and editing are different

Carrier screening

Tests prospective parents for inherited risks.

PGT-M

Uses IVF to test embryos for a known single-gene condition.

Polygenic screening

Uses many variants to estimate complex disease risk.

Genome editing

Changes DNA; heritable changes may reach future generations.

A simplified PGT-M pathway

Genetic counselling ➡ **IVF** ➡ **Embryo testing** ➡ **Transfer decision**

Selection chooses among embryos; it does not repair their DNA.

c. Research paper

Lencz and colleagues • eLife

“Utility of polygenic embryo screening for disease depends on the selection strategy”

Aim

Find out how the embryo selection strategy changes the predicted risk of a complex disease.

Method

Used statistical models and simulated offspring based on real genetic data.

Data used

Used genetic data from schizophrenia and Crohn’s disease studies to check the model.

What was compared?

Excluding very high-risk embryos versus choosing the embryo with the lowest risk score.

A polygenic score uses many gene variants to estimate disease risk.

c. Findings of the research paper

The selection strategy changed the predicted benefit

Exclude very high-risk embryos

Removing only embryos with very high scores gave relatively small predicted risk reductions.

Choose the lowest-risk embryo

Choosing the lowest-score embryo could give much larger relative risk reductions, if enough viable embryos were available.

Why did the results vary?

The benefit changed with score accuracy, embryo number, disease frequency, and the parents' genetic risks.

A relative risk reduction does not mean that the disease risk becomes zero.

c. Limitations and meaning of the paper

What we can and cannot conclude

- This was a modelling study, not a trial following children after embryo screening.
- Predictions depended on accurate risk scores and enough viable embryos.
- The model focused on one disease at a time; results for individual couples could differ.

What this means for our topic

The way embryos are ranked matters. But this paper does not prove that screening guarantees a healthy child or that every use is ethically acceptable.

d. Benefits and limitations

Possible benefits depend on the practice and the evidence

Possible benefits

- Helps families understand inherited risks.
- May reduce the chance of a serious single-gene condition.
- Can give families more reproductive choices.
- Can support informed planning and counselling.

Limitations

- Many traits depend on genes and environment.
- Prediction does not guarantee health.
- IVF can be costly and physically demanding.
- Few embryos, test errors, and unequal access can limit benefit.

Strongest case: voluntary, reliable testing for a well-understood serious inherited condition.

e. Ethical implications

Scientific ability does not automatically make an action right

- Consent: People must be free to accept or refuse testing.
- Disability rights: A condition does not reduce a person’s worth.
- Fairness: Expensive tests may increase inequality.
- Privacy: Genetic information also affects relatives.
- Pressure: Families must not be blamed for declining selection.
- Embryos: People hold different views about their moral status.

Forced sterilization and reproductive discrimination violate basic rights.

Sources: [3], [4], [6]

e. When could selective reproduction be acceptable?

A simple way to think about the question

Reliable evidence

Does the test predict a clearly defined condition?

➡

Free choice

Can the family refuse without threats or penalties?

➡

Fair safeguards

Are benefits, burdens, rights, and alternatives explained?

Voluntary PGT-M for a serious inherited disorder	May be justified after counselling and careful assessment.
Forced sterilization or racial selection	Ethically unacceptable.
Promises to create a “perfect” child	Not supported by genetic prediction.

Sources: [3], [4], [6]

f. Future perspective

Better evidence and stronger safeguards are both needed

- Develop more accurate genetic tests for different populations.
- Study actual health outcomes, not only predicted scores.
- Explain risks in clear language, including absolute risk.
- Improve counselling, privacy protection, and fair access.
- Use strong oversight for heritable genome editing.
- Include people with genetic conditions in policy discussions.

Improve treatment and support as well as prevention.

g. Conclusion

A balanced answer to the topic

Scientifically

Using tests to guide embryo selection can reduce some inherited disease risks. Complex traits remain hard to predict. No test guarantees health.

Ethically

Voluntary choices may be acceptable with good evidence, counselling, and safeguards. Coercion and discrimination are not acceptable.

Selective reproduction may sometimes be justified for a serious inherited condition—but never as a way to rank human worth.

h. References

One research paper and supporting sources

[1] Lencz, T., Backenroth, D., Granot-Hershkovitz, E., et al. Utility of polygenic embryo screening for disease depends on the selection strategy. eLife, 10, e64716.

DOI: 10.7554/eLife.64716

[2] GeneReviews®. HEXA Disorders. NCBI Bookshelf.

Open source

[3] Human Fertilisation and Embryology Authority. PGT-M and PGT-SR: patient information.

Open source

[4] World Health Organization. Human Genome Editing: Recommendations.

Open source

[5] United States Holocaust Memorial Museum. Eugenics. Holocaust Encyclopedia.

Open source

[6] United Nations. Convention on the Rights of Persons with Disabilities, Article 23.

Open source